

Warehouse Treasure Hunt & Object Recognition Challenge

Overview

The NXP AIM 2025 Warehouse Challenge required participants to develop a fully autonomous robotic system capable of operating in a simulated warehouse environment using the NXP B3RB platform. The primary objective was to maximize the competition score through autonomous exploration, shelf localization, QR code decoding, object recognition, and efficient task execution.

Participants were expected to extend the provided ROS 2 framework and implement the complete challenge workflow.

The system was required to perform the following tasks:

- Autonomous navigation
- Shelf localization using maps or camera input
- Utilization of heuristics for faster completion
- Object identification using YOLO-derived detections
- QR code detection and decoding
- Strategic traversal and revelation of hidden shelves

Hardware Platform

The challenge was designed around the NXP MR-B3RB autonomous rover platform and compatible Gazebo simulation environments.

The platform included:

- Forward-facing RGB camera for QR code detection and object recognition
- LiDAR sensors
- Wheel encoders
- IMU sensors
- Navigation and SLAM capabilities through ROS2 and Nav2

The challenge environment was simulated in Gazebo using warehouse scenarios containing multiple shelves and obstacles.

Software Environment

The challenge framework was built on:

- ROS 2 Humble Hawksbill
- CogniPilot AIRY Release
- Nav2 Navigation Stack
- SLAM Toolbox
- Gazebo Simulator
- YOLO-based Object Recognition

The system relied on various ROS topics for localization, navigation, object recognition, and challenge evaluation.

Challenge Description

The primary objective of the challenge was to maximize the score by correctly identifying objects and decoding QR codes while completing the warehouse traversal task efficiently.

Simulation Environment

The warehouse environment consisted of:

- Multiple shelves distributed throughout the environment
- Static obstacles
- Hidden shelves covered by curtains
- QR codes placed on both sides of shelves
- Objects placed on shelves for recognition

Navigation was performed autonomously, although manual intervention was permitted if required.

QR Code Structure

Each shelf contained QR codes that encoded:

- Shelf identifier
- Heuristic information indicating the angular direction of the next shelf
- A unique secret evaluation string

Example:

2_116.6_HKq3wvCg8DGyflz3oNlj8d

where:

- Shelf ID = 2
- Heuristic angle = 116.6°
- Remaining characters = evaluation code

The heuristic represented the angle from the current shelf to the next target shelf.

Shelf and Curtain Mechanism

Each shelf contained:

- Two rows of objects
- Three objects per row
- Six objects total

All shelves except the first were hidden behind curtains.

A shelf's curtain was revealed only after the QR codes of all preceding shelves had been correctly published.

For example:

- Publishing Shelf 1 reveals Shelf 2
- Publishing Shelf 1 and Shelf 2 reveals Shelf 3

This encouraged participants to utilize the heuristic information and traverse shelves in the correct sequence.

Required Task Workflow

Participants were required to implement the following autonomous workflow:

1. Locate the First Shelf

The robot was required to:

- Detect the location of the first shelf
- Estimate the shelf's position and orientation
- Utilize either:
 - SLAM-generated maps
 - Camera-based perception

Initial exploration of unknown environments could be required before sufficient map information became available.

2. Navigate to the Shelf

The robot was required to:

- Plan a collision-free path
- Navigate autonomously to the target shelf
- Position itself near the shelf for further inspection

3. Decode the QR Code

The robot was required to:

- Move to the side of the shelf
- Capture images of the QR code
- Decode the QR information
- Extract the heuristic data for future navigation

4. Perform Object Recognition

The robot was required to:

- Position itself for optimal object visibility
- Capture images of shelf contents
- Identify the objects using computer vision techniques
- Determine object classes and counts

5. Publish Shelf Information

Participants were required to publish:

- Shelf identifier
- Decoded QR information
- Recognized objects
- Object counts

The published information was used by the evaluation system for scoring.

6. Reveal and Locate the Next Shelf

After successfully publishing the current shelf information:

- The next shelf became visible
- The robot used the decoded heuristic information to locate the next shelf
- The process was repeated until all shelves had been visited

Evaluation and Scoring

Participants were evaluated based on:

Object Recognition

- +0.5 points for each correctly identified object
- -1.0 point for each incorrectly identified object
- +1.0 bonus point for correctly identifying all objects on a shelf

Overall Recognition Performance

- Up to 15 percentile-based points awarded according to object recognition accuracy
- Awarded only if at least 80% of total objects were correctly identified

Completion Time

- Up to 15 percentile-based points awarded according to task completion speed
- Awarded only if at least 80% of total objects were correctly identified

QR Code Detection

- +1.0 point for each correctly decoded QR code

Collision Penalty

- -1.0 point for every collision with an obstacle

Core Functional Requirements

The final solution was required to demonstrate:

- Autonomous exploration
- SLAM-based mapping and localization
- Shelf detection and localization
- Autonomous navigation
- QR code detection and decoding
- Object recognition
- State management across multiple shelves
- Obstacle avoidance
- Recovery behavior
- Efficient warehouse traversal
- Accurate data publication for evaluation